M^3UCD: A Multi-task Multimodal Metaphor Understanding Challenge Dataset for LLMs

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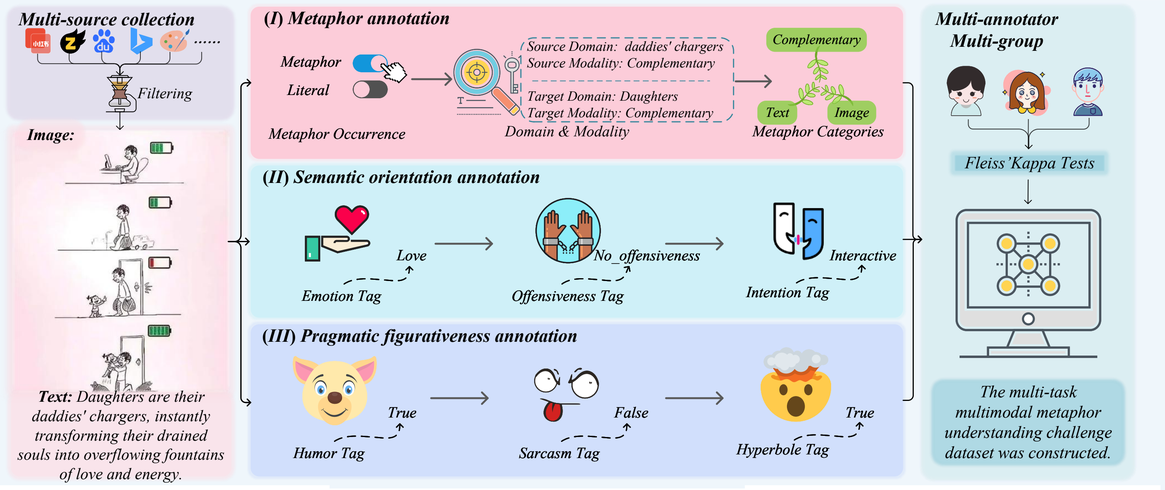
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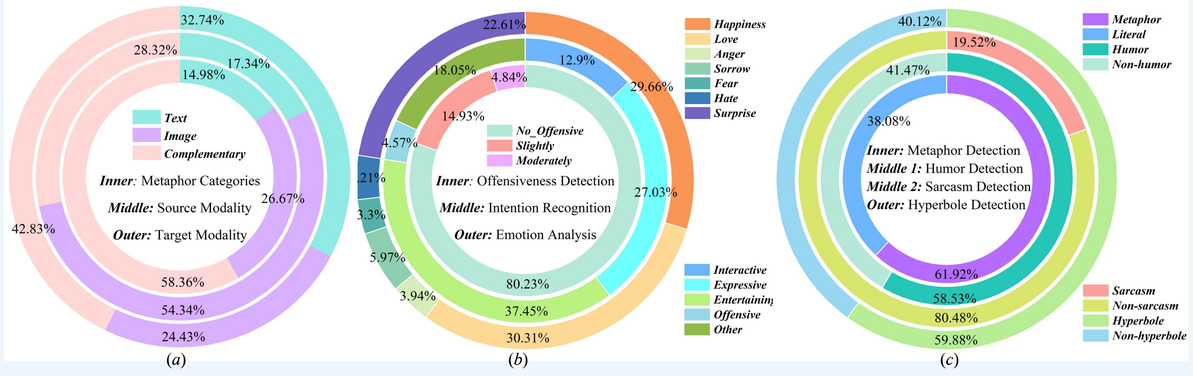
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Introduction

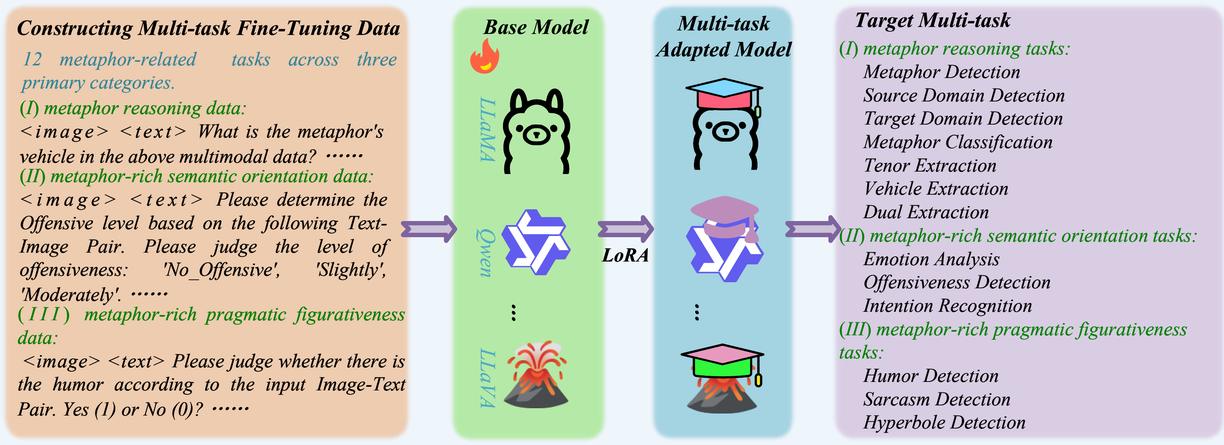
- ***Problem*** (1) ***Datasets with limited annotation dimensions***, lacking deep semantic orientation and pragmatic figurativeness annotations related to metaphors, which hinders fine-grained metaphor analysis. (2) ***Absence of systematic evaluation on LLMs' performance in multimodal metaphor understanding (MMU) tasks***, coupled with the lack of fine-grained evaluation benchmarks for assessing LLMs in this domain. (3) ***Imperfect task construction for MMU***, where current research remains confined to basic detection tasks without delving into semantic orientation and pragmatic figurativeness in metaphorical mapping processes, nor establishing standardized paradigms for critical subtasks like target/source domain extraction (tenor/vehicle extraction).
- ***Methodology*** (1) ***Dataset***: We present the largest and most richly annotated fine-grained multi-task MMU dataset, featuring manual annotations across multiple dimensions: metaphor occurrence/categories, semantic orientation, pragmatic figurativeness, and tenors/vehicles. Rigorous quality control through annotation guidelines and inter-annotator agreement analysis ensures dataset reliability. (2) ***LLMs Benchmarking***: We conduct systematic evaluation of LLMs' capabilities on fine-grained metaphor understanding. Benchmark results across various LLMs demonstrate M^3UCD's potential and generalizability, establishing valuable references for future MMU research. (3) ***Task and Framework Formulation***: Beyond extending semantic orientation and pragmatic figurativeness analysis tasks inherent to metaphor-rich, we formalize the novel task of tenor/vehicle extraction in MMU and introduce a unified Multi-task Collaborative Learning Framework (MCLF) to enhance the metaphor understanding of MLLMs.

The annotation pipeline with the M^3UCD





Multi-task Collaborative Learning Framework (MCLF)



We define a multi-task learning setup comprising T metaphor-related tasks. Let $\mathcal{D}^{(t)} = \{(x_i, p_i^{(t)}, y_i^{(t)})\}_{i=1}^{N_t}$ be the data corresponding to task t, where $x_i = (I_i, T_i)$ denotes a multimodal input consisting of an image-text pair, $p_i^{(t)}$ is the natural language prompt guiding task t, and $y_i^{(t)}$ is its corresponding label or target. Each task uses a head $f^{(t)}_{\theta+\Delta\theta}$ on top of a frozen base MLLM with parameters θ; only the low-rank adaptation parameters $\Delta\theta$ (LoRA) are trainable.

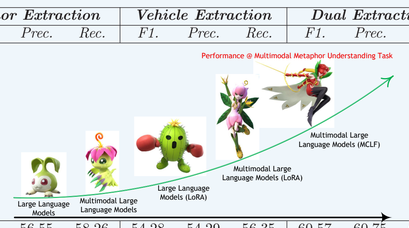
$$\mathcal{L}_{\text{total}} = \sum_{t=1}^{T} \lambda_t \, \mathbb{E}_{\substack{(x_i,\, p_i^{(t)},\, y_i^{(t)}) \\ \sim \mathcal{D}^{(t)}}} \left[\mathcal{L}^{(t)}\big(f^{(t)}_{\theta+\Delta\theta}(x_i,\, p_i^{(t)}),\, y_i^{(t)} \big) \right] \tag{1}$$

where $\lambda_t \equiv 1$, $\mathcal{L}^{(t)}$ denotes the loss function associated with task t. We adopt per-task losses with a piecewise definition; classification tasks use cross-entropy and generation tasks use a masked autoregressive language-model loss.

The MCLF is trained by optimizing the total loss with respect to the LoRA parameters, i.e., $\min_{\Delta\theta} \mathcal{L}_{\text{total}}$, while keeping the base model parameters θ frozen. The framework supports a unified optimization strategy for both classification and generation-based MMU tasks across multi-task setting.

Experiments

Type	*Model*	*Metaphor-rich Semantic Orientation Tasks*						*Metaphor-rich Pragmatic Figurativeness Tasks*						*Metaphor Reasoning Tasks*																
		EA		*OD*		*IR*		*HumorD*		*SD*		*HyperD*		*MD*		*SDD*		*TDD*		*MC*		*Tenor Extraction*			*Vehicle Extraction*			*Dual Extraction*		
		Acc.	*F1.*	*Acc.*	*F1.*	*Acc.*	*F1.*	*Acc.*	*F1.*	*Acc.*	*F1.*	*Acc.*	*F1.*	*Acc.*	*F1.*	*Acc.*	*F1.*	*Acc.*	*F1.*	*Acc.*	*F1.*	*F1.*	*Prec.*	*Rec.*	*F1.*	*Prec.*	*Rec.*	*F1.*	*Prec.*	*Rec.*
	Random	14.04	11.30	33.52	25.39	19.35	17.33	50.85	50.79	50.98	44.99	50.29	49.89	50.49	49.56	30.65	29.15	31.79	31.59	34.98	32.42									
PTMs	Bi-LSTM	48.53	41.97	82.80	47.67	53.03	46.04	72.64	70.09	87.88	77.60	76.29	74.91	74.14	69.83	63.73	56.38	53.35	52.57	65.17	58.14									
	BERT	51.07	43.92	84.63	59.46	57.46	51.34	76.38	75.60	88.57	80.79	79.09	77.68	78.47	76.99	68.63	61.28	58.88	57.87	67.49	63.81									
	ResNet50	40.39	28.94	75.34	48.43	40.26	34.77	69.32	67.79	80.46	71.18	77.49	76.15	69.77	67.08	53.30	47.85	47.01	44.92	56.50	47.23									
	ViT	37.79	20.10	81.14	42.15	41.40	29.11	70.55	67.07	80.91	68.69	74.40	73.89	70.68	68.38	57.07	45.90	45.36	42.76	58.31	49.33									
	MAE	37.39	24.96	70.09	43.52	42.21	28.82	69.54	65.27	80.62	66.79	73.71	72.79	70.29	65.56	56.86	48.67	46.39	43.12	60.63	43.84									
	DeiT	41.34	30.97	80.26	51.65	45.18	35.06	70.98	70.25	84.63	68.52	74.04	73.81	72.31	69.16	55.37	49.00	50.00	47.89	60.22	49.95									
	SwinT	40.20	26.19	80.94	48.33	44.07	32.78	71.73	70.42	82.54	73.14	76.09	75.42	70.52	69.74	60.58	49.60	50.05	43.19	62.28	49.73									
	CLIP	53.49	44.69	84.98	60.45	59.77	54.20	76.78	75.97	89.64	82.78	79.58	78.66	77.20	75.29	66.77	62.44	58.93	58.77	67.65	63.15									
	FLAVA	38.53	23.11	82.67	43.10	46.22	29.01	67.59	66.24	83.94	69.38	70.68	68.26	69.87	66.75	56.19	36.65	47.47	42.63	59.55	24.88									
LLMs	ChatGLM3-6B-Base	30.39	11.57	73.88	35.67	16.19	12.83	60.00	52.64	47.46	44.64	55.57	50.32	61.92	46.05	20.59	14.49	32.30	19.85	17.65	14.16	56.32	56.55	58.26	54.28	54.29	56.35	60.57	60.75	61.14
	LLaMA3-8B-Instruct	22.38	15.97	69.54	40.20	28.66	11.67	60.33	54.84	54.69	49.79	53.97	52.10	60.52	50.81	21.21	20.02	34.93	28.26	33.28	25.86	50.06	46.55	55.56	51.63	49.50	55.26	56.80	54.48	59.76
	InternLM2.5-7B-Chat	35.21	19.95	79.28	41.50	26.84	21.06	57.23	56.80	67.04	56.29	53.45	53.02	58.66	54.28	18.68	12.08	32.97	18.74	15.94	10.02	54.31	50.73	59.89	52.54	49.40	57.61	59.94	57.53	63.13
	LLaMA3.1-8B-Instruct	30.62	10.59	71.07	33.73	14.95	7.83	58.89	42.36	21.89	20.48	58.60	41.93	61.76	42.91	19.56	14.60	33.38	21.78	18.94	13.64	45.48	39.31	55.90	45.73	39.85	55.79	53.53	48.76	59.93
	MiniCPM3-4B-Chat	31.43	15.84	76.51	45.38	27.23	23.90	49.38	48.26	78.93	60.43	48.11	45.44	57.39	54.89	18.68	11.55	32.97	17.43	15.58	9.48	57.82	62.04	55.36	57.72	62.82	54.78	61.12	64.68	58.31
	GPT-3.5-turbo	37.07	27.57	70.29	39.98	24.20	21.68	57.17	56.95	63.19	52.09	48.73	45.55	59.74	57.44	18.47	11.52	33.13	17.53	15.38	8.96	57.35	58.41	57.49	54.66	55.24	55.39	60.77	62.57	59.51
MLLMs	InternVL2-1B	32.77	19.09	79.22	34.48	14.36	11.13	38.79	31.12	81.34	44.85	40.26	28.20	36.84	33.02	27.30	14.80	25.28	13.53	25.44	14.12	53.08	48.80	59.72	53.68	50.67	58.83	60.60	57.89	64.10
	InternVL2-2B	33.62	21.14	35.80	29.43	33.00	25.49	58.11	50.59	81.89	54.03	49.87	45.91	52.87	52.25	51.29	31.49	25.44	13.59	51.96	31.94	55.80	53.36	60.18	55.00	53.72	58.14	61.38	61.09	62.31
	InternVL2-4B	35.21	23.43	79.54	45.47	33.71	22.50	59.35	59.31	23.29	22.01	58.96	49.47	63.42	39.88	54.08	26.18	40.20	22.30	58.46	27.26	53.98	49.54	60.77	52.64	48.95	58.43	60.13	58.39	62.47
	LLaVA1.5-7B-Chat	17.98	11.78	46.58	32.11	29.09	19.09	58.37	41.11	36.22	35.83	52.44	50.50	60.65	43.07	18.68	14.75	35.29	24.09	20.95	15.86	55.27	54.38	57.92	55.61	64.36	50.99	61.11	62.09	60.74
	Qwen2-VL-7B-Instruct	29.80	17.85	72.44	37.95	37.23	18.19	54.07	53.59	75.73	59.24	47.17	44.64	52.25	51.85	39.22	33.24	39.16	34.28	46.13	35.54	59.40	61.78	58.68	58.77	63.12	56.54	63.57	66.65	61.32
	InternVL2-8B	38.86	27.41	80.72	30.40	39.25	28.60	61.24	61.20	76.86	69.17	57.72	56.94	63.39	50.69	52.53	30.81	26.16	17.27	38.85	34.53	56.00	52.98	60.84	54.01	51.96	57.69	61.33	60.78	62.44
	Llama-3.2-11B-Vision-Instruct	31.60	21.09	75.02	47.48	35.90	23.58	51.99	51.18	64.95	56.85	54.10	54.03	63.16	39.29	35.91	31.78	37.41	33.01	54.54	30.05	54.68	55.32	56.23	53.72	49.05	61.24	61.46	58.27	65.70
	InternVL2-26B	39.71	30.50	81.07	35.09	34.46	28.84	70.13	67.66	77.82	70.22	66.94	65.98	66.32	49.94	34.98	25.24	42.26	32.62	59.13	29.36	57.02	54.50	61.27	55.85	54.09	59.22	62.92	62.24	64.23
	InternVL2-40B	39.61	30.78	80.23	43.61	38.60	27.15	70.39	69.78	71.07	66.33	60.00	58.93	65.96	48.93	49.48	34.37	44.84	31.73	58.00	35.51	58.73	56.83	62.15	59.15	59.99	59.75	64.40	64.87	64.49
	Qwen2-VL-72B-Instruct	37.70	25.32	80.84	41.76	42.03	21.25	67.74	67.06	82.70	58.57	55.29	54.56	66.89	53.64	34.59	30.37	33.76	32.62	59.27	26.89	60.19	61.61	60.14	59.01	60.56	59.00	66.44	67.63	65.84
	InternVL2-Llama3-76B	37.56	27.04	65.83	42.28	34.40	25.91	59.15	38.04	50.62	48.71	59.22	37.71	61.50	48.82	34.11	24.09	43.45	31.29	53.87	32.45	57.54	57.71	59.50	54.48	55.63	55.37	61.17	62.69	60.47
	GLM-4V-Plus	35.86	20.36	82.21	47.80	34.04	26.36	61.43	59.73	75.73	67.81	55.31	55.21	63.13	38.78	19.31	12.39	32.58	25.69	24.88	19.26	58.48	60.05	58.67	55.55	58.10	54.76	62.56	64.09	61.66
	GPT-4o	37.65	26.65	82.18	48.74	37.72	19.56	71.01	70.78	81.92	72.72	67.30	58.30	71.27	62.32	30.08	21.76	42.52	24.43	58.77	26.49	60.91	62.49	60.70	60.05	62.77	58.81	66.30	68.28	64.92
	Gemini-1.5 Pro	39.54	29.44	81.04	35.52	41.14	21.16	64.43	64.21	84.50	77.33	64.85	55.97	66.94	51.42	48.56	42.24	39.63	35.44	59.34	28.29	60.67	61.49	61.28	61.85	62.16	62.94	66.71	67.00	66.93
	Claude-3.5-Sonnet	37.92	28.26	80.78	38.48	40.00	25.54	63.55	63.43	83.81	76.97	66.35	64.92	65.57	47.13	47.24	39.50	40.22	35.27	56.93	36.23	58.00	56.97	60.46	57.50	56.71	59.79	63.97	63.21	65.23
	o3-mini	40.29	29.71	80.72	32.79	36.35	26.26	58.86	57.30	83.13	58.65	59.74	59.64	67.65	63.34	46.96	41.97	39.63	36.62	55.26	41.44	59.70	60.35	60.43	60.39	62.03	60.32	65.35	66.29	64.98
	Claude-3-7-Sonnet-Thinking	39.09	28.75	80.78	38.27	40.59	24.46	63.16	62.67	83.65	76.59	67.72	65.63	67.26	52.26	46.54	36.44	40.71	33.00	58.15	34.04	57.78	57.18	59.72	58.22	57.87	60.05	64.25	63.88	65.09
	GPT-4.1	38.37	25.74	81.89	40.48	29.54	11.30	66.55	66.54	78.24	64.26	66.51	55.29	66.68	49.43	39.83	31.10	42.47	26.55	59.29	28.10	58.83	58.57	60.40	60.14	60.53	61.14	66.36	66.61	66.63
	Gemini-2.5-flash-preview	40.52	26.25	80.55	30.81	37.56	23.17	68.01	67.97	83.68	67.05	72.41	70.35	68.93	57.53	39.32	37.63	42.11	33.24	60.06	31.73	60.60	60.00	62.62	60.16	60.06	61.70	66.62	66.25	67.53
LLMs (LoRA)	ChatGLM3-6B-Base (LoRA)	42.28	35.53	84.14	57.92	53.00	46.22	68.14	66.44	88.27	80.34	70.78	69.01	76.06	74.12	55.57	45.93	43.29	40.63	55.01	44.57	60.55	62.94	59.83	59.69	62.82	58.39	63.93	66.00	62.47
	LLaMA3-8B-Instruct (LoRA)	46.12	38.82	84.10	57.33	53.32	46.02	75.47	74.53	89.38	82.12	74.01	72.52	73.68	71.56	60.22	51.76	46.85	45.46	57.84	47.91	61.31	64.26	60.26	60.14	63.49	58.80	64.52	66.77	62.97
	InternLM2.5-7B-Chat (LoRA)	42.38	35.02	83.65	52.51	48.34	39.26	67.30	65.46	88.83	81.07	70.85	69.47	70.20	67.75	45.30	42.40	45.30	42.40	54.54	44.31	61.09	63.68	60.38	59.54	62.28	58.77	64.27	66.29	62.95
	LLaMA3.1-8B-Instruct (LoRA)	43.26	37.26	83.36	53.35	46.64	39.15	71.37	69.99	89.15	81.52	69.09	67.42	68.31	65.72	53.77	39.45	42.83	40.25	51.50	40.05	60.47	62.98	59.79	59.32	62.40	58.27	64.03	66.03	62.67
	MiniCPM3-4B-Chat (LoRA)	39.84	32.71	82.90	51.51	46.03	38.57	68.11	65.54	87.98	79.72	69.41	67.49	67.52	63.75	52.63	41.32	42.47	39.85	54.23	41.78	61.20	64.31	59.70	58.95	62.62	57.28	64.15	66.92	62.07
MLLMs (LoRA)	LLaVA1.5-7B-Chat (LoRA)	42.54	36.49	84.27	55.34	47.95	39.95	74.20	72.90	90.07	82.98	77.23	76.02	74.66	72.43	56.97	48.00	43.70	42.54	54.33	44.58	59.83	62.47	58.96	59.35	62.65	58.05	64.08	66.11	62.67
	LLaVA-1.5-13B-Chat (LoRA)	53.42	46.61	86.51	63.80	60.07	54.26	80.39	79.69	91.17	85.59	80.94	80.24	79.15	77.48	62.80	56.13	52.68	51.79	61.40	56.06	64.46	66.22	64.27	62.99	65.14	62.59	67.75	68.92	67.13
	LLaVA-NeXT-7B-Chat (LoRA)	45.44	38.97	84.30	56.14	49.25	44.19	78.70	77.87	91.04	85.02	79.51	78.40	76.25	74.27	57.12	49.58	43.55	41.67	59.34	51.99	61.76	64.13	61.01	60.24	63.11	59.30	65.01	66.79	63.90
	LLaVA-NeXT-13B-Chat (LoRA)	54.01	47.49	86.22	64.70	57.65	52.35	80.23	79.46	91.37	85.79	81.82	81.11	80.26	78.72	64.04	57.60	51.75	51.05	61.82	55.74	65.15	67.10	64.80	63.49	65.84	62.87	68.21	69.51	67.50
	Qwen2-VL-7B-Instruct (LoRA)	42.15	36.32	84.07	57.73	52.35	44.86	78.99	78.34	91.40	85.88	81.14	80.17	80.13	78.25	62.69	55.36	49.07	46.56	61.51	54.44	65.14	67.66	64.34	64.62	68.08	63.06	68.62	70.94	67.03
	Qwen2.5-VL-3B-Instruct (LoRA)	43.55	36.65	84.01	57.29	51.82	45.00	76.91	75.99	90.85	85.17	82.31	81.59	79.67	78.20	64.96	59.00	53.51	52.57	62.95	56.72	65.34	67.61	64.83	65.33	68.07	64.47	69.26	70.92	68.25
	Qwen2.5-VL-7B-Instruct (LoRA)	53.62	48.60	86.45	64.69	59.48	53.96	81.60	80.92	92.35	87.27	82.74	82.04	80.98	79.68	63.05	56.61	50.05	48.60	60.63	53.03	65.83	68.22	65.12	65.83	69.01	64.50	69.63	71.69	68.24
	Llama-3.2-11B-Vision-Instruct (LoRA)	48.44	42.21	85.08	59.63	54.36	48.99	79.80	79.09	90.49	84.70	80.26	79.56	76.78	75.48	64.81	58.18	49.95	48.33	64.50	55.36	64.14	66.78	63.26	64.53	67.58	63.37	68.32	70.12	67.19
MLLMs (MCLF)	LLaVA1.5-7B-Chat (MCLF)	54.14	47.57	83.45	61.47	58.01	52.22	80.42	79.60	90.78	84.93	80.36	79.55	78.31	76.60	68.01	62.51	54.02	53.70	66.15	61.61	64.87	66.31	64.97	64.12	65.96	63.88	68.26	69.31	67.76
	LLaVA-1.5-13B-Chat (MCLF)	53.94	47.71	85.44	60.10	59.38	53.84	80.39	79.60	90.68	84.58	81.37	80.48	78.86	77.15	68.32	62.93	55.16	54.87	67.91	63.49	66.05	67.34	66.18	64.75	66.15	64.89	69.06	69.88	68.80
	LLaVA-NeXT-7B-Chat (MCLF)	52.80	45.52	85.21	61.07	57.49	51.66	79.41	78.51	90.36	84.01	81.34	80.59	77.75	76.04	68.11	63.33	53.72	53.24	67.29	62.78	64.59	66.22	64.51	64.23	66.02	64.02	68.34	69.36	67.87
	LLaVA-NeXT-13B-Chat (MCLF)	54.98	47.77	86.61	64.13	60.16	54.79	79.90	78.87	91.27	85.64	81.56	80.75	80.16	78.52	68.42	63.32	55.37	55.12	66.41	62.00	66.73	68.00	66.88	64.90	66.35	64.90	69.49	70.19	69.49
	Qwen2-VL-7B-Instruct (MCLF)	54.36	48.29	86.74	63.99	58.86	52.63	81.63	80.90	92.28	87.30	83.49	82.73	80.94	79.45	68.73	64.73	56.55	55.92	66.20	62.64	68.33	69.86	68.38	67.18	68.75	67.14	71.13	72.06	70.81
	Qwen2.5-VL-3B-Instruct (MCLF)	51.92	45.56	85.90	62.11	56.42	51.49	81.01	80.31	90.75	84.91	80.65	79.94	77.17	75.48	65.89	60.28	52.48	51.68	63.98	59.33	66.23	67.98	66.12	65.66	67.78	65.23	69.86	71.12	69.22
	Qwen2.5-VL-7B-Instruct (MCLF)	53.97	46.08	86.48	63.99	59.22	53.31	82.28	81.55	91.79	86.70	82.57	81.78	80.07	78.50	69.09	64.45	56.09	55.34	66.31	61.96	67.28	68.51	67.59	66.94	68.53	66.99	70.66	71.42	70.49
	Llama-3.2-11B-Vision-Instruct (MCLF)	54.63	48.09	86.25	62.84	59.64	53.09	80.72	79.83	91.76	86.36	81.56	80.80	79.45	77.99	67.08	61.74	59.49	58.60	68.73	63.87	67.32	69.00	67.20	66.39	68.14	66.26	70.44	71.40	70.05
	Avg. Improvement of MCLF over LoRA	+5.95	+5.41	+0.65	+2.55	+4.53	+4.93	+1.87	+1.86	+0.12	+0.25	+0.87	+0.94	+0.60	+0.65	+5.90	+7.85	+6.08	+6.92	+5.81	+8.72	+2.47	+1.63	+3.16	+2.22	+1.03	+3.14	+2.05	+1.22	+2.82



Experimental results of PTMs, LLMs, and MLLMs on metaphor-rich semantic orientation, metaphor-rich pragmatic figurativeness, and metaphor reasoning tasks. Cyan means the second best. Purple means the best. Green indicates the average improvement of MCLF over LoRA.

Conclusion

1) Our research established the largest fine-grained multi-task multimodal metaphor understanding dataset (M^3UCD) to date. This dataset comprises 15,345 multi-label annotation instances.

2) We conducted systematic evaluations of advanced PTMs, LLMs and MLLMs on this annotated content. Additionally, we developed a concise and unified MCLF for metaphor understanding.

3) M^3UCD's complexity presents significant challenges for MMU tasks, with even cutting-edge models like GPT-4o, Claude-3-7-Sonnet-Thinking struggling with metaphor understanding. MCLF effectively enhances MLLMs' metaphor understanding. This underscores the necessity for developing novel metaphor understanding paradigms.